

SofiBuddy: A Soft Mobile Interface for On-Body Interaction

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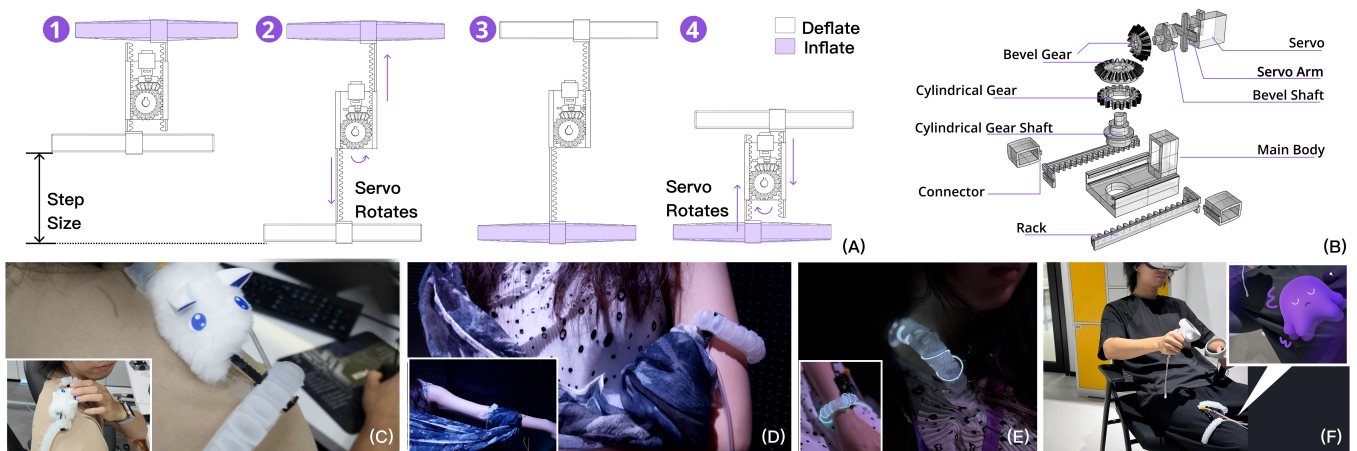


Figure 1: (A) Overview of the SofiBuddy. (B) Exploded view of the mechanical module. Application scenarios: (C) Emotional Companion, (D) Expressive Wearable Ornament, (E) Light Feedback for Stress Relief, and (F) Immersive VR Experiences.

Abstract

On-body robots offer new forms of direct, expressive interaction with the human body, yet many existing designs rely on rigid structures that limit comfort and tactility. We present SofiBuddy, a soft mobile interface with a lightweight transition module and a curvature-based inflatable actuator, offering gentle bodyV conforming motion with interactive potential for companionship, assistance, and VR engagement.

CCS Concepts

• **Human-centered computing** → *Interaction design*;

Keywords

Soft interface, On-body robot, Mobile interaction, Interaction design

ACM Reference Format:

Yixuan Li, Jie Yu, Chenwan Halley Zhong, Zhaowen Deng, Teng Han, Yi Cai, and Xin Tong. 2025. SofiBuddy: A Soft Mobile Interface for On-Body Interaction. In *SIGGRAPH Asia 2025 Posters (SA Posters '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3757374.3771432>

1 INTRODUCTION

On-body robots have gained attention as wearable devices capable of directly interacting with the human body, with applications in healthcare [Dementyev et al. 2018], companionship [Li et al. 2024], and interactive wearables such as moving jewelry [Dementyev et al. 2016] and microphone interfaces [Kim et al. 2022]. The proximity of these systems to the body enables intimate, expressive interactions,

fostering potential for rich human–robot relationships. However, their physical integration demands high comfort and seamless bodily connection for user acceptance. While prior work has explored attachment strategies such as suction-based adhesion [Dementyev et al. 2018], magnetic wheels [Dementyev et al. 2016], and 3D-printed tracks [Sathya et al. 2022], these strategies often lead to attachment discomfort and unnatural bodily sensations [Sathya et al. 2022]. Given these limitations, soft interfaces offer unique advantages in on-body robotic interaction over traditional rigid systems due to their compliance, deformability, and biological resemblance. Their inherent softness and mechanical adaptability enable safe, appealing, and socially acceptable interactions with humans [Jørgensen et al. 2022]. These interfaces can offer new ways of engaging human perception, especially in scenarios that require direct and dynamic feedback to the body [He et al. 2015].

We present *SofiBuddy*, a soft mobile interface for on-body interaction that combines a lightweight transition module with a curvature-based inflatable actuator. Through an exploratory study and four application scenarios, we highlight the potential of soft interfaces to enhance interactions in daily life.

2 DESIGN AND IMPLEMENTATION

SofiBuddy consists of a lightweight mechanical module, two soft actuators, and a compact control system.

Transition Mechanism. We designed a 10.2 g mechanical module powered by a 4.3 g servo motor for compact on-body use. The motor drives bevel gears that convert rotational motion into linear displacement via a rack-and-pinion system (Fig. 1.A, B), with a 1.43:1 torque-increasing gear ratio. The base embedded in silicone (Ecoflex 00-30) ensures comfortable skin contact during motion.

Soft Actuator. Based on the bending model [Polygerinos et al. 2013], we design our actuator to maximize curvature through material and geometric tuning. To optimize bending and enhance conformity to human limb shapes, we fabricated the prototype with Ecoflex 30, reducing t while preserving a relatively high b (Fig. 2.A). We fabricated the prototype using Ecoflex 30, reducing

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SA Posters '25, Hong Kong, Hong Kong

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ACM ISBN 979-8-4007-2134-2/25/12

<https://doi.org/10.1145/3757374.3771432>

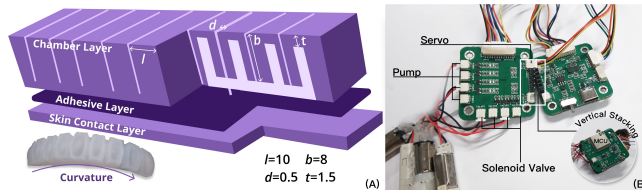


Figure 2: (A) Soft actuator structure with inflatable bending design. (B) Compact control system overview.

wall thickness t while maintaining a chamber height b to enhance bending and better conform to the shape of human limbs (Fig. 2.A).

Control System. We developed a lightweight control system on a 3×3 cm PCB, optimized for hardware minimization and real-time actuation, which can be worn at the waist or placed in a pocket for mobility (Fig. 2.B). An ESP32-C3 MCU coordinates a servo motor, two miniature diaphragm pumps (SC3101PM, Skookum Electronic Co.), and two solenoid valves (S070C-SDG-32, SMC Pneumatics).

3 EXPLORATORY STUDY AND RESULTS

Study Design. We conducted an exploratory study with six HCI researchers (age range: 19-26, mean age: 23.7, 5 female). The study focused on two aspects: users' perceptions of SofiBuddy's on-body interaction and ideations of future use cases.

Results. Participants described *SofiBuddy* as “soothing and natural” ($n=4$), and “skin-friendly” ($n=5$), noting that its gentle movement and presence felt intuitive for on-body interaction. The actuator was perceived as emotionally expressive, resembling “gentle massage” ($n=3$). The system was seen as providing pet-like interactions ($n=4$), with tactile feedback valued as a subtle, less disruptive alternative to visual or audio cues.

Application Scenarios. Suggested use cases included companionship, stress relief, social communication, and ambient reminders, with desired improvements in controllable behavior (e.g., GUI) and varied actuation (e.g., tapping, grasping). Based on study findings, we present four conceptual application scenarios to explore on-body interactions. *SofiBuddy* is a soft wearable that acts as an emotional companion through gentle, pet-like touch (Fig.1.C), and as a dynamic ornament that combines expressive motion with utility, such as adjusting clothing (Fig.1.D). Its actuators support synchronized light and rhythmic inflation for calming stress relief (Fig.1.E), while embedded sensors enable responsive feedback. In VR, it serves as a tactile proxy for virtual creatures, translating actions into real skin sensations to deepen immersion (Fig.1.F).

Acknowledgments

We thank the Guangzhou Science and Technology Bureau for their support through the Guangzhou City-University Joint Research Grant (Project No. GP2025M024, Grant No. P00437).

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